

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1 FT	(a)	(i)		1.6 [cm]		1		1		
		(ii)		0.5 [cm] ignore minus sign		1		1		
	(b)	(i)		5 Hz Accept any means of identifying the correct answer		1		1	1	
		(ii)		Selection of $v = f\lambda$ or by implication (1) Substitution: $v = 5 \times 1.6$ (1) ecf on f and λ $v = 8$ [cm/s] (1) Alternative: Selection of $\text{speed} = \frac{\text{distance}}{\text{time}}$ or by implication of any $\frac{\text{distance}}{2}$ (1) Substitution: $v = \frac{10 \times 1.6}{2}$ ecf (1) $v = 8$ [cm/s] (1) Answer of $v = 8$ [cm/s] not necessarily on the answer line award 3 marks	1 1	1		3	2	
	(c)			Wavefronts at 90° by eye at least 3 but all drawn with a ruler (1) Constant wavelength same as the incident wavelength for all gaps by eye (1) N.B. Wavefronts must be drawn on reflected ray to access any marks		2		2		
				Question 1 total	2	6	0	8	3	0